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LECTURER: DR. MUGOYE

Assignment Goals

In this assignment, you will explore a real-world Information Systems problem within the Kenyan context something that genuinely affects an organization, sector, or community you are familiar with or interested in. You are expected to:

i. Clearly describe the issue and its context and explain why the problem is significant. (3 marks)
ii. Select and justify a philosophical stance (e.g., positivism, interpretivism, or critical theory) and explain how this stance shapes your understanding of the problem. (4 marks)
iii. Identify at least two relevant Information Systems theories or frameworks and explain how they are relevant to the problem. (3 marks)
iv. Reflect on why you chose this problem and perspective, linking to your research interests or professional experience. (3 marks)
v. Oral Défense, deliver a 5-minute presentation defending your philosophical and theoretical choices.

Fragmented Agricultural Data Management and Knowledge Sharing Systems at KALRO: An Information Systems Analysis

Abstract

This report examines a critical Information Systems problem within the Kenya Agricultural and Livestock Research Organization (KALRO): the fragmentation of agricultural data management and knowledge sharing systems that impedes effective research dissemination and extension service delivery. Despite KALRO's mandate to generate and disseminate agricultural research for national food security, the organization faces significant challenges in managing research data, integrating digital platforms, and facilitating bidirectional information flow between researchers, extension officers, and farmers. This problem is analyzed through a positivist philosophical lens, applying the Technology Acceptance Model (TAM) and the DeLone and McLean Information Systems Success Model to understand adoption barriers and system effectiveness. The analysis reveals that inadequate data curation infrastructure, poor system interoperability, limited user involvement in system design, and insufficient digital literacy among stakeholders collectively undermine KALRO's capacity to fulfill its mission. Addressing these challenges requires strategic investments in integrated information systems, capacity building, and participatory design approaches that prioritize end-user needs and contextual realities of Kenyan agriculture.

1. Introduction

Agricultural research organizations in developing countries face unprecedented challenges in managing and disseminating knowledge in an increasingly digital world. The Kenya Agricultural and Livestock Research Organization (KALRO), as Kenya's premier national agricultural research institution, stands at the intersection of traditional agricultural extension and digital transformation. However, the organization confronts a critical Information Systems (IS) problem: fragmented data management and knowledge sharing systems that impede its capacity to effectively generate, curate, and disseminate agricultural research to diverse stakeholders including extension officers, farmers, policymakers, and the broader agricultural community.

This report provides a comprehensive analysis of this IS problem, examining its manifestations, underlying causes, and implications for KALRO's mission. The analysis is grounded in empirical

evidence from recent scholarly literature on agricultural information systems in Kenya and employs a positivist philosophical approach to understand the problem through measurable constructs and testable relationships. Two prominent IS theories,the Technology Acceptance Model (TAM) and the DeLone and McLean IS Success Model,are applied to illuminate the factors influencing system adoption and effectiveness within KALRO's organizational context.

The significance of this problem extends beyond KALRO's institutional boundaries. As Kenya pursues agricultural transformation to achieve food security, climate resilience, and economic development, the effectiveness of agricultural research dissemination becomes paramount. Fragmented IS infrastructure not only limits KALRO's operational efficiency but also constrains the translation of research findings into practical agricultural innovations that can benefit millions of smallholder farmers across Kenya.

2. Background and Context

2.1 KALRO's Mandate and Organizational Context

The Kenya Agricultural and Livestock Research Organization (KALRO) was established through the Science, Technology and Innovation Act of 2013 as a merger of 15 agricultural research institutes. KALRO's mandate encompasses conducting agricultural research, promoting research findings and technologies, and providing agricultural information services to support Kenya's agricultural sector. The organization operates across diverse agro-ecological zones, conducting research on crops, livestock, food systems, and natural resource management.

KALRO's organizational structure includes research centers distributed across Kenya, each focusing on specific commodities or thematic areas. The organization employs hundreds of researchers, technical staff, and support personnel who generate substantial volumes of research data, publications, and agricultural innovations annually. However, the institutional merger that created KALRO brought together disparate information systems, data management practices, and organizational cultures, creating integration challenges that persist today.

2.2 The Digital Transformation Imperative in Agricultural Research

Global trends in agricultural research emphasize the importance of digital transformation, open data, and knowledge management systems. International agricultural research organizations have increasingly adopted digital platforms for data management, collaborative research, and

stakeholder engagement. In Kenya, the government's digital economy strategy and the broader push for e-government services create expectations for KALRO to modernize its information systems and enhance digital service delivery.

However, digital transformation in agricultural research organizations faces unique challenges in developing country contexts. These include limited ICT infrastructure in rural areas, varying levels of digital literacy among stakeholders, resource constraints, and the need to balance technological innovation with contextual appropriateness. Research indicates that many ICT for agriculture (ICT4Ag) initiatives in Kenya have struggled with adoption and sustainability, often due to inadequate attention to user needs, contextual factors, and system integration [1].

3. The Information Systems Problem: Description and Significance

3.1 Problem Description

The core IS problem at KALRO can be characterized as fragmented agricultural data management and knowledge sharing systems that fail to support effective research dissemination and extension service delivery. This problem manifests across multiple dimensions of KALRO's information infrastructure:

Data Management Fragmentation: KALRO's research data exists in disparate formats and locations, with limited standardization, metadata documentation, or centralized curation. Research data curation at KALRO's libraries faces significant challenges including inadequate infrastructure, limited technical expertise, insufficient funding, and lack of institutional policies for research data management [2]. This fragmentation impedes data discovery, reuse, and integration across research projects and centers.

System Interoperability Deficiencies: Agricultural information systems deployed within KALRO and across Kenya's agricultural sector often operate as isolated silos with limited interoperability. Many ICT4Ag initiatives are generalized rather than contextualized, facilitating one-way information push to farmers rather than enabling bidirectional information flow [3]. This lack of integration prevents seamless data exchange between research management systems, extension platforms, and farmer-facing applications.

Extension Service Digitalization Gaps: KALRO's extension services, which bridge research and practice, face significant digitalization challenges. Extension officers have not fully adopted

current information access and dissemination technologies, remaining dependent on traditional methods requiring physical movement [4]. E-extension platforms deployed in Kenya suffer from poor usability, insufficient user consultation during design, technical language barriers, and inadequate coverage of diverse crops and livestock [5].

Knowledge Management Deficiencies: Despite generating substantial research outputs, KALRO lacks comprehensive knowledge management strategies to capture, organize, and disseminate institutional knowledge effectively. Knowledge management in East African agricultural research organizations faces challenges including limited ICT infrastructure, inadequate funding, and insufficient staff training [6].

3.2 Manifestations of the Problem

The fragmented IS infrastructure at KALRO manifests in several observable ways:

1. Limited Research Data Accessibility: Research data generated by KALRO scientists often remains inaccessible to other researchers, extension officers, and external stakeholders due to poor documentation, lack of digital repositories, and absence of data sharing policies [2].
2. Inefficient Extension Service Delivery: Extension officers struggle to access timely, relevant agricultural information from KALRO's research outputs, limiting their capacity to provide evidence-based advisory services to farmers [5].
3. Duplication of Efforts: Without integrated systems for tracking research activities and outputs, KALRO faces risks of duplicating research efforts across centers and missing opportunities for collaborative research.
4. Weak Feedback Mechanisms: The absence of bidirectional information flow prevents KALRO researchers from receiving systematic feedback from extension officers and farmers about research priorities, technology performance, and emerging agricultural challenges [3].
5. Digital Divide Challenges: KALRO's digital transformation initiatives face adoption barriers related to digital literacy, infrastructure limitations, and inadequate consideration of diverse user contexts [1].

3.3 Significance of the Problem

This IS problem is significant for multiple reasons:

Mission-Critical Impact: KALRO's core mandate,generating and disseminating agricultural research,depends fundamentally on effective information systems. Fragmented IS infrastructure directly undermines the organization's capacity to fulfill its national role in agricultural development.

Food Security Implications: Kenya faces persistent food security challenges exacerbated by climate change, population growth, and land degradation. Effective dissemination of agricultural research and technologies is essential for improving agricultural productivity and resilience. IS fragmentation creates barriers between research innovations and farmer adoption, limiting the potential impact of KALRO's research on national food security.

Resource Efficiency: KALRO operates with limited financial and human resources. Fragmented IS infrastructure leads to inefficiencies through duplicated data entry, redundant systems, and time-consuming manual processes. Integrated systems could enhance operational efficiency and allow researchers to focus on core scientific activities.

Accountability and Impact Assessment: Donors, government, and other stakeholders increasingly demand evidence of research impact and accountability for public investments in agricultural research. Fragmented data systems impede KALRO's capacity to systematically track research outputs, monitor technology adoption, and demonstrate impact.

Digital Economy Participation: As Kenya advances its digital economy agenda, KALRO's capacity to participate in digital agricultural ecosystems, contribute to open data initiatives, and collaborate with international research networks depends on modernizing its IS infrastructure.

Equity and Inclusion: Fragmented and poorly designed IS systems can exacerbate existing inequalities in access to agricultural information. Research on digital agricultural innovations in Kenya highlights challenges in operationalizing inclusivity, with technology adoption barriers disproportionately affecting marginalized farmers [1].

4. Philosophical Stance: Positivism

4.1 Selection and Justification

This analysis adopts a **positivist philosophical stance** to examine KALRO's IS problem. Positivism, rooted in the natural sciences, holds that reality is objective and can be observed and

measured through empirical methods. In IS research, positivism emphasizes the identification of causal relationships between measurable variables, hypothesis testing, and the development of generalizable knowledge through quantitative methods.

The selection of positivism is justified by several considerations:

Alignment with Problem Nature: The IS problem at KALRO involves measurable phenomena,system usage rates, data quality metrics, user adoption patterns, and information flow efficiency,that lend themselves to positivist inquiry. Understanding these phenomena requires systematic measurement and analysis of relationships between variables such as perceived usefulness, system quality, and user satisfaction.

Theory Testing Orientation: This analysis applies established IS theories (TAM and DeLone-McLean IS Success Model) that were developed within positivist traditions and are designed for hypothesis testing and empirical validation. These theories specify measurable constructs and testable relationships that align with positivist epistemology.

Practical Decision-Making Support: Positivist research produces actionable findings that can inform management decisions and system design. For KALRO's leadership, understanding which factors significantly influence system adoption or which system quality dimensions most strongly predict user satisfaction provides concrete guidance for resource allocation and intervention design.

Complementarity with Existing Research: The majority of empirical research on technology acceptance in agricultural contexts employs positivist approaches, utilizing surveys, structural equation modeling, and hypothesis testing [7], [8], [9]. Adopting positivism allows this analysis to build upon and integrate findings from this established research base.

4.2 How Positivism Shapes Understanding of the Problem

The positivist stance shapes this analysis in several important ways:

Focus on Measurable Constructs: Positivism directs attention to observable and quantifiable aspects of the IS problem. Rather than focusing solely on subjective interpretations or social constructions of technology, the analysis examines measurable factors such as system response time, data accuracy, user training hours, and adoption rates. This focus enables systematic comparison across different systems, user groups, and organizational contexts.

Causal Explanation: Positivism seeks to identify causal relationships between variables. In analyzing KALRO's IS problem, this translates to examining questions such as: Does perceived

ease of use causally influence system adoption? Does information quality affect user satisfaction? Does user satisfaction lead to increased system use? Understanding these causal relationships enables prediction and intervention design.

Generalizability: Positivist research aims to develop generalizable knowledge that extends beyond specific cases. While this analysis focuses on KALRO, the positivist approach seeks to identify patterns and relationships that may apply to other agricultural research organizations in similar contexts. This generalizability enhances the broader contribution of the analysis to IS scholarship and practice.

Hypothesis-Driven Inquiry: Positivism emphasizes hypothesis formulation and testing. The application of TAM and the DeLone-McLean model involves specifying hypotheses about relationships between constructs (e.g., "perceived usefulness positively influences behavioral intention to use") and examining whether empirical evidence supports these hypotheses in KALRO's context.

Objectivity and Replicability: Positivism values researcher objectivity and study replicability. This analysis strives to base conclusions on empirical evidence from published research rather than subjective impressions, and to specify methods and reasoning transparently so that others could replicate or extend the analysis.

Limitations Acknowledged: While positivism offers valuable strengths, it is important to acknowledge its limitations. Positivist approaches may underemphasize the social, political, and cultural dimensions of IS problems. Technology adoption is not purely a rational process driven by measurable utility; it is also shaped by organizational politics, power dynamics, cultural values, and historical contexts that are difficult to quantify. A purely positivist analysis might miss important insights about how KALRO's organizational culture, power structures, or historical legacy influence IS adoption and effectiveness. Complementary interpretivist or critical approaches could provide richer understanding of these dimensions. However, for the purposes of this analysis,identifying measurable factors influencing IS effectiveness and adoption,positivism provides an appropriate and productive philosophical foundation.

5. Theoretical Frameworks

5.1 Technology Acceptance Model (TAM)

5.1.1 Overview of TAM

The Technology Acceptance Model, developed by Fred Davis in 1989, is one of the most widely applied theories in IS research for understanding and predicting user acceptance of information technologies. TAM posits that two primary beliefs,Perceived Usefulness (PU) and Perceived Ease of Use (PEOU),determine an individual's attitude toward using a technology, which in turn influences behavioral intention and actual system use.

Perceived Usefulness is defined as "the degree to which a person believes that using a particular system would enhance his or her job performance." Perceived Ease of Use refers to "the degree to which a person believes that using a particular system would be free of effort." TAM theorizes that PEOU influences PU (easier systems are perceived as more useful) and that both constructs influence attitude toward use, behavioral intention, and ultimately actual usage behavior.

TAM has been extensively validated across diverse technologies and contexts, demonstrating robust predictive power. The model's parsimony and strong empirical support have made it a foundational framework in IS research. Numerous extensions of TAM have incorporated additional variables such as social influence, facilitating conditions, self-efficacy, and trust to enhance explanatory power in specific contexts [7], [8], [10].

5.1.2 Relevance to KALRO's IS Problem

TAM is highly relevant to understanding KALRO's IS problem for several reasons:

Explaining Adoption Barriers: KALRO's fragmented IS infrastructure includes multiple systems and platforms that have experienced low adoption rates among researchers, extension officers, and farmers. TAM provides a framework for systematically examining why these systems are underutilized. If users perceive systems as difficult to use (low PEOU) or not useful for their work (low PU), adoption will be limited regardless of system technical capabilities.

Empirical Evidence from Agricultural Contexts: Extensive research has applied TAM to agricultural technology adoption, providing empirical evidence of its validity in contexts similar to KALRO. Studies of agricultural information systems, e-learning platforms, precision agriculture technologies, and mobile agricultural services consistently find that perceived

usefulness and perceived ease of use significantly influence adoption intentions [7], [8], [11], [12], [13].

Identifying Intervention Points: TAM's specification of key constructs provides clear targets for intervention. If PEOU is low, interventions might focus on user training, interface redesign, or technical support. If PU is low, interventions might emphasize demonstrating system benefits, improving system functionality, or better aligning system features with user needs.

User-Centered Design Implications: TAM emphasizes that technology acceptance depends on user perceptions, not just objective system characteristics. This aligns with evidence that many ICT4Ag initiatives in Kenya have failed due to inadequate user consultation during design [5]. TAM reinforces the importance of participatory design approaches that prioritize user needs and perceptions.

Extension to KALRO Context: TAM can be extended to incorporate contextual factors relevant to KALRO. Research on agricultural technology adoption has successfully extended TAM to include variables such as facilitating conditions (infrastructure, technical support), social influence (peer adoption, extension officer recommendations), self-efficacy (confidence in using technology), and job relevance (alignment with work tasks) [7], [8], [14]. These extensions enhance TAM's applicability to KALRO's complex organizational and technological environment.

5.1.3 Application to KALRO's Specific Challenges

Applying TAM to KALRO's IS challenges yields several insights:

Data Management Systems: KALRO's research data management systems likely suffer from low perceived usefulness if researchers do not see clear benefits for their work (e.g., if data sharing is not rewarded or if systems do not facilitate data analysis). Low perceived ease of use may result from complex interfaces, inadequate training, or technical difficulties. Addressing these perceptions requires both improving system design and demonstrating tangible benefits of data management.

Extension Platforms: E-extension platforms face adoption challenges among both extension officers and farmers. For extension officers, platforms may have low PU if they do not provide timely, relevant information or if they do not support officers' workflow. For farmers, platforms may have low PEOU due to digital literacy barriers, language issues, or poor mobile network

connectivity [5]. TAM suggests that successful extension platforms must address both usefulness (providing valuable agricultural information) and ease of use (simple interfaces, local language support, offline functionality).

Researcher Information Systems: Systems for managing research projects, publications, and collaborations may have low adoption if researchers perceive them as administrative burdens (low PU) or as complex and time-consuming (low PEOU). Enhancing adoption requires demonstrating how these systems support research productivity and simplifying user interactions.

5.2 DeLone and McLean Information Systems Success Model

5.2.1 Overview of the Model

The DeLone and McLean IS Success Model, first proposed in 1992 and updated in 2003, provides a comprehensive framework for conceptualizing and measuring IS effectiveness. The updated model specifies six interrelated dimensions of IS success:

1. **System Quality**: Technical characteristics such as reliability, response time, ease of use, and functionality.
2. **Information Quality**: Characteristics of system outputs including accuracy, completeness, currency, and relevance.
3. **Service Quality**: Quality of support provided to users, including responsiveness, empathy, and assurance.
4. **Use/Intention to Use**: Extent and nature of system utilization.
5. **User Satisfaction**: Users' overall satisfaction with the system.
6. **Net Benefits**: Positive impacts of the system on individuals, organizations, or society.

The model posits that system quality, information quality, and service quality influence both use and user satisfaction, which in turn affect net benefits. The model also includes feedback loops, recognizing that positive experiences and benefits reinforce continued use and satisfaction.

The DeLone and McLean model has been widely applied and validated across diverse IS contexts, from enterprise systems to e-commerce to e-government. Its multidimensional conceptualization of IS success provides a holistic framework for evaluating system effectiveness beyond simple usage metrics.

5.2.2 Relevance to KALRO's IS Problem

The DeLone and McLean IS Success Model is particularly relevant to KALRO's IS problem for several reasons:

Comprehensive Evaluation Framework: KALRO's IS challenges are multifaceted, involving technical system issues, data quality problems, inadequate user support, and limited organizational benefits. The DeLone and McLean model provides a comprehensive framework for diagnosing problems across all these dimensions rather than focusing narrowly on single aspects.

Information Quality Focus: A central aspect of KALRO's IS problem is poor information quality,research data that is incomplete, poorly documented, not standardized, or inaccessible [2]. The model explicitly incorporates information quality as a critical dimension of IS success, providing a framework for understanding how data quality deficiencies undermine system effectiveness.

Service Quality Emphasis: KALRO's IS challenges include inadequate technical support, limited training, and insufficient user assistance [5], [6]. The model's inclusion of service quality recognizes that IS success depends not only on system and data quality but also on the support infrastructure that enables effective use.

Net Benefits Assessment: The ultimate goal of KALRO's IS infrastructure is to generate benefits,improved research productivity, enhanced extension service delivery, better agricultural outcomes. The model's focus on net benefits directs attention to whether systems actually deliver value rather than simply measuring usage or satisfaction.

Diagnostic Capability: The model's specification of relationships between dimensions provides diagnostic insights. For example, if KALRO's systems have low usage despite adequate system quality, the problem may lie in information quality, service quality, or failure to demonstrate net benefits. This diagnostic capability helps prioritize interventions.

5.2.3 Application to KALRO's Specific Challenges

Applying the DeLone and McLean model to KALRO's IS challenges reveals several insights:

System Quality Deficiencies: Evidence suggests that KALRO's IS infrastructure suffers from system quality problems including poor interoperability, inadequate functionality for bidirectional information flow, and usability issues [3], [5]. These system quality deficiencies directly undermine use and user satisfaction. Addressing them requires investments in system integration, user interface improvements, and functionality enhancements aligned with user needs.

Information Quality Problems: KALRO's research data management challenges reflect fundamental information quality deficiencies,incomplete metadata, lack of standardization, poor documentation, and limited data validation [2]. These information quality problems reduce the usefulness of data repositories and research databases, limiting their adoption and impact. Improving information quality requires establishing data standards, implementing quality control processes, and building data curation capacity.

Service Quality Gaps: KALRO's IS infrastructure appears to suffer from service quality gaps including inadequate user training, limited technical support, and insufficient user involvement in system design [5], [6]. These service quality deficiencies leave users struggling to effectively utilize systems, reducing satisfaction and use. Addressing service quality requires investments in training programs, help desk services, and participatory design processes.

Limited Net Benefits: The fragmented nature of KALRO's IS infrastructure limits the net benefits realized by the organization. Researchers face inefficiencies in data management, extension officers struggle to access research findings, and farmers receive limited benefit from digital extension platforms. Demonstrating and enhancing net benefits requires not only improving system, information, and service quality but also better communicating the value of IS investments to stakeholders.

Feedback Loops: The model's feedback loops are particularly relevant to KALRO's situation. Negative experiences with poorly designed systems, low-quality data, or inadequate support create vicious cycles where users avoid systems, reducing the data and feedback needed to improve them. Breaking these negative cycles requires interventions that improve user experiences and demonstrate tangible benefits, creating positive feedback loops that reinforce adoption and continuous improvement.

6. Analysis and Discussion

6.1 Technology Adoption Barriers at KALRO

Applying TAM to KALRO's context reveals multiple barriers to technology adoption across different user groups:

Perceived Usefulness Barriers: Research indicates that many agricultural information systems in Kenya suffer from low perceived usefulness due to several factors. Systems often provide generalized information rather than contextualized, actionable guidance relevant to specific

farming contexts [3]. Extension platforms frequently lack information on specific crops or livestock relevant to particular regions [5]. For KALRO researchers, data management systems may not clearly demonstrate benefits for research productivity or career advancement. These perceived usefulness deficiencies directly undermine adoption intentions.

Perceived Ease of Use Barriers: Multiple studies document ease of use challenges in Kenyan agricultural IS. E-extension platforms suffer from poor usability, technical language that is difficult for farmers to understand, and interfaces not designed for users with limited digital literacy [5]. Poor mobile network coverage in rural areas creates technical difficulties that reduce perceived ease of use [4]. Inadequate training and technical support leave users struggling to effectively operate systems [6]. These ease of use barriers are particularly significant given that TAM research consistently shows PEOU is especially important for novel technologies where users lack experience [15].

Facilitating Conditions: Extended TAM models incorporate facilitating conditions,organizational and technical infrastructure that supports technology use. KALRO faces significant facilitating conditions challenges including limited ICT infrastructure, inadequate funding for IS investments, and insufficient technical personnel [2], [6]. These deficiencies in facilitating conditions constrain adoption even when users perceive systems as useful and easy to use.

Social Influence: Research on agricultural technology adoption emphasizes the importance of social influence,the degree to which individuals perceive that important others believe they should use a technology [14], [16]. In KALRO's context, social influence operates at multiple levels. Extension officers' adoption of digital platforms is influenced by peer adoption and organizational expectations. Farmers' adoption is influenced by extension officer recommendations and peer farmer experiences. Building positive social influence requires demonstrating successful use cases and leveraging trusted intermediaries.

Self-Efficacy: Computer self-efficacy,confidence in one's ability to use technology,significantly influences technology acceptance in agricultural contexts [8]. KALRO's diverse user base includes individuals with varying levels of digital literacy and technology experience. Low self-efficacy among some researchers, extension officers, and farmers creates psychological barriers to adoption. Building self-efficacy requires not only training but also designing systems that provide positive early experiences and build user confidence.

6.2 System Quality and Information Quality Deficiencies

Applying the DeLone and McLean model highlights critical quality deficiencies in KALRO's IS infrastructure:

System Quality Issues: KALRO's IS infrastructure exhibits multiple system quality problems. Poor interoperability between systems creates data silos and prevents integrated workflows [3]. Many systems facilitate only one-way information push rather than bidirectional communication [3]. Usability problems make systems difficult to navigate and use effectively [5]. These system quality deficiencies directly reduce user satisfaction and limit system use.

Information Quality Challenges: Information quality problems are particularly acute in KALRO's research data management. Research data often lacks adequate metadata, making it difficult to discover and interpret [2]. Data standardization is limited, impeding integration across projects and centers. Data completeness and accuracy are inconsistent due to inadequate quality control processes. These information quality deficiencies undermine the value of data repositories and limit data reuse, reducing the return on research investments.

Service Quality Deficiencies: KALRO's IS infrastructure suffers from service quality gaps. User training is inadequate, leaving many users unable to effectively utilize systems [6]. Technical support is limited, forcing users to struggle with problems independently. Most critically, users are often not consulted during system design, resulting in systems that do not meet actual user needs [5]. These service quality deficiencies create frustration and reduce user satisfaction.

Interrelationships: The DeLone and McLean model emphasizes that quality dimensions are interrelated. Poor system quality (e.g., slow response times) can undermine perceived information quality even if data is accurate. Inadequate service quality (e.g., lack of training) can make even well-designed systems difficult to use. KALRO's IS challenges reflect problems across all three quality dimensions, requiring comprehensive interventions rather than narrow technical fixes.

6.3 Organizational and Contextual Factors

Beyond the factors specified in TAM and the DeLone-McLean model, several organizational and contextual factors shape KALRO's IS challenges:

Organizational Integration Challenges: KALRO was formed through the merger of 15 previously independent research institutes, each with its own information systems, data management practices, and organizational culture. This legacy creates integration challenges as

the organization attempts to develop unified IS infrastructure. Resistance to change, competing priorities across centers, and lack of standardized processes impede IS integration efforts.

Resource Constraints: KALRO operates with limited financial and human resources. Inadequate funding constrains investments in IS infrastructure, training, and technical support [2], [6]. Limited numbers of ICT personnel restrict the organization's capacity to develop, maintain, and support complex information systems. These resource constraints force difficult trade-offs and limit the pace of IS modernization.

Digital Divide: Kenya's digital divide,disparities in access to ICT infrastructure, digital literacy, and connectivity between urban and rural areas,directly affects KALRO's IS initiatives. Many of KALRO's research centers operate in rural areas with limited internet connectivity. Extension officers serve farmers in remote locations with poor mobile network coverage [4]. These infrastructure limitations constrain the types of IS solutions that are feasible and effective.

Policy and Governance Gaps: KALRO lacks comprehensive policies for research data management, data sharing, and IS governance [2]. This policy vacuum creates uncertainty about data ownership, access rights, and responsibilities. Without clear policies, researchers may be reluctant to share data, and the organization struggles to enforce data management standards.

Capacity and Skills Gaps: KALRO faces capacity gaps in data management, information systems, and digital literacy across its workforce [2], [6]. Many researchers lack training in research data management best practices. Extension officers have limited experience with digital extension platforms. These capacity gaps limit the organization's ability to effectively utilize IS infrastructure even when it is available.

Inclusivity Challenges: Research on digital agricultural innovations in Kenya highlights challenges in operationalizing inclusivity [1]. Technology adoption barriers disproportionately affect marginalized groups including women farmers, elderly farmers, and farmers in remote areas. KALRO's IS initiatives must address these equity dimensions to ensure that digital transformation does not exacerbate existing inequalities in access to agricultural knowledge and services.

7. Implications and Recommendations

7.1 Strategic Recommendations

Based on the analysis, several strategic recommendations emerge for addressing KALRO's IS challenges:

1. Adopt Integrated IS Architecture: KALRO should develop and implement an integrated IS architecture that enables interoperability across research management, data repositories, extension platforms, and farmer-facing applications. This architecture should be based on open standards, APIs, and data exchange protocols that allow different systems to communicate seamlessly. Integration will reduce data silos, eliminate redundant data entry, and enable comprehensive workflows from research generation to farmer adoption.

2. Implement Comprehensive Research Data Management Program: KALRO should establish a comprehensive research data management program including data policies, standards, repositories, and curation services. This program should specify requirements for metadata documentation, data quality control, and data sharing. Investing in data curation capacity at KALRO libraries will enhance data discoverability, reusability, and long-term preservation [2].

3. Prioritize User-Centered Design: All IS development and procurement should employ participatory, user-centered design approaches that actively involve end users,researchers, extension officers, and farmers,throughout the design process. This approach will ensure that systems address actual user needs, workflows, and contexts rather than imposing generic solutions [3], [5]. User involvement should extend beyond initial design to include iterative testing, feedback collection, and continuous improvement.

4. Invest in Digital Capacity Building: KALRO should implement comprehensive capacity building programs targeting digital literacy, data management skills, and IS competencies across its workforce. Training should be tailored to different user groups and should emphasize practical skills and hands-on experience. Building internal capacity will enhance the organization's ability to effectively utilize IS infrastructure and reduce dependence on external technical support [6].

5. Enhance Service Quality Infrastructure: KALRO should invest in service quality infrastructure including help desk services, technical support, user documentation, and training programs. Adequate service quality is essential for supporting users, resolving problems quickly, and building user confidence and satisfaction. Service quality investments should be viewed as integral to IS success rather than optional add-ons.

6. Demonstrate and Communicate Net Benefits: KALRO should systematically document and communicate the benefits of IS investments to stakeholders. This includes demonstrating how data management systems enhance research productivity, how extension platforms improve service delivery, and how integrated systems generate organizational efficiencies. Communicating net benefits will build support for continued IS investments and motivate user adoption.

7. Address Facilitating Conditions: KALRO should work with government, donors, and partners to address facilitating conditions including ICT infrastructure, connectivity, and resource availability. This may involve advocating for improved rural internet connectivity, securing dedicated funding for IS investments, and building partnerships with ICT providers to enhance infrastructure access.

8. Develop Contextualized Solutions: IS solutions should be contextualized to Kenya's agricultural realities rather than simply adopting generic international platforms. Contextualization includes supporting local languages, accommodating limited connectivity through offline functionality, addressing diverse crop and livestock systems, and aligning with local agricultural practices and knowledge systems [3].

7.2 Policy Implications

The analysis also yields several policy implications:

National Agricultural Research Policy: Kenya's agricultural research policy should explicitly address IS infrastructure, data management, and digital transformation as strategic priorities. Policy should mandate research data management, support open data initiatives, and provide dedicated funding for IS investments in agricultural research organizations.

Digital Agriculture Strategy: Kenya's digital agriculture strategy should recognize the critical role of research organizations like KALRO in generating and disseminating agricultural knowledge. Strategy should address the integration of research systems with extension platforms and farmer-facing applications to create seamless knowledge flows from research to practice.

Capacity Building Policy: Agricultural education and training policy should incorporate digital literacy, data management, and IS competencies as core components of curricula for agricultural researchers, extension officers, and agricultural students. Building these competencies at scale requires systemic changes in agricultural education.

Infrastructure Investment: Government ICT infrastructure policy should prioritize rural connectivity to support digital agricultural services. Improved connectivity is a prerequisite for effective digital extension, precision agriculture, and data-driven farming.

Open Data and Knowledge Sharing: Government policy should promote open data and knowledge sharing in agricultural research while addressing legitimate concerns about data ownership, intellectual property, and commercial interests. Clear policies can reduce uncertainty and encourage data sharing that benefits the broader agricultural sector.

8. Reflection on Problem and Perspective Selection

The selection of KALRO's fragmented data management and knowledge sharing systems as the focus of this analysis reflects both research interests and professional considerations. As agricultural research organizations globally confront digital transformation imperatives, understanding the IS challenges they face and identifying evidence-based solutions becomes increasingly important. KALRO provides a compelling case because it represents a large, complex agricultural research organization operating in a developing country context where IS challenges are particularly acute yet where effective agricultural research dissemination is most critical for food security and development.

The choice to analyze this problem through a positivist lens, applying TAM and the DeLone-McLean IS Success Model, reflects a pragmatic orientation toward generating actionable insights that can inform management decisions and system design. While interpretivist or critical approaches could provide valuable complementary insights,for example, examining how power dynamics shape IS adoption or how different stakeholders construct meanings around technology,the positivist approach offers clear advantages for identifying measurable factors that influence IS effectiveness and for developing testable interventions.

The analysis reveals that KALRO's IS challenges are not primarily technical problems requiring only better technology. Rather, they are socio-technical problems involving complex interactions between technology, people, organizations, and contexts. Effective solutions require not only technical improvements but also organizational change, capacity building, policy development, and attention to user needs and contextual realities. This insight reinforces the value of IS research that examines technology in organizational and social contexts rather than treating technology as isolated artifacts.

The analysis also highlights the importance of evidence-based approaches to IS challenges in developing country contexts. Too often, IS initiatives in these contexts are driven by technology enthusiasm or donor priorities rather than systematic understanding of user needs, organizational capabilities, and contextual constraints. Applying established IS theories like TAM and the DeLone-McLean model to KALRO's context demonstrates how theoretical frameworks can guide empirical investigation and generate insights that inform more effective interventions.

Finally, this analysis underscores the broader significance of IS research for development. Information systems are not neutral tools but rather interventions that can either reinforce or challenge existing inequalities, power structures, and development trajectories. KALRO's IS challenges have implications not only for organizational efficiency but also for food security, agricultural sustainability, and equitable access to agricultural knowledge. Addressing these challenges effectively requires IS research and practice that is attentive to development goals, equity considerations, and the specific needs and contexts of developing countries.

9. Conclusion

This report has examined a critical Information Systems problem within the Kenya Agricultural and Livestock Research Organization: fragmented agricultural data management and knowledge sharing systems that impede effective research dissemination and extension service delivery. Through a positivist analytical lens, applying the Technology Acceptance Model and the DeLone and McLean IS Success Model, the analysis has identified multiple dimensions of this problem including technology adoption barriers, system quality deficiencies, information quality challenges, service quality gaps, and organizational and contextual constraints.

The analysis reveals that KALRO's IS challenges are fundamentally socio-technical in nature, involving complex interactions between technology characteristics, user perceptions and behaviors, organizational factors, and contextual realities. Effective solutions require comprehensive interventions addressing system design, data quality, user support, capacity building, policy development, and organizational change. Narrow technical fixes will be insufficient; what is needed is a holistic approach that recognizes the multidimensional nature of IS success.

The application of TAM and the DeLone-McLean model demonstrates the value of established IS theories for understanding and addressing real-world problems in developing country contexts.

These theories provide frameworks for systematically diagnosing problems, identifying intervention points, and predicting the likely effects of different solutions. However, the analysis also reveals the importance of contextualizing these theories to specific organizational and national contexts, recognizing that factors such as resource constraints, infrastructure limitations, and digital divides shape IS challenges and solutions in distinctive ways.

Ultimately, addressing KALRO's IS challenges is not merely an organizational imperative but a development priority. Effective agricultural research dissemination is essential for Kenya's food security, agricultural transformation, and sustainable development. By modernizing its IS infrastructure, adopting user-centered design approaches, building digital capacity, and implementing evidence-based interventions, KALRO can enhance its capacity to fulfill its national mandate and contribute to agricultural development. The insights and recommendations from this analysis provide a foundation for this transformation, grounded in empirical evidence and established IS theory.

10. References

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